

2022 Ph H2 Q8

Section: Particles and Waves

Topic: Radiation & Spectra

Light from sodium lamp studied with collimator and screen. (a)(i) Calculate power from irradiance and spot size. (a)(ii) Explain why setup unsuitable for inverse square law. (b) Spectroscope: (i) electron transition direction, (ii) energy difference for 589.0 nm line, (iii) explain relative brightness of 589.0 nm vs 589.6 nm lines.

Worked solution

(a)(i) Power of beam

$$I = P/A \Rightarrow P = I \cdot A$$

$$A = \pi r^2 = \pi(0.015)^2 = 7.07 \times 10^{-4} \text{ m}^2$$

$$P = 17 \times 7.07 \times 10^{-4} = 0.012 \text{ W}$$

Answer: $1.21 \times 10^{-2} \text{ W}$

(a)(ii) Inverse square law

For inverse square, irradiance should decrease as distance increases because energy spreads over a larger area. But here, the spot radius is fixed by the collimator, so area is constant, irradiance stays constant. Hence unsuitable for testing inverse square law.

(b)(i) Light emission means electrons drop to a lower energy level.

(b)(ii) Energy difference

$$E = hc/\lambda$$

$$= (6.63 \times 10^{-34})(3.0 \times 10^8)/(589.0 \times 10^{-9})$$

$$= 3.38 \times 10^{-19} \text{ J}$$

$$\text{Convert to eV: } \div 1.60 \times 10^{-19} = 2.11 \text{ eV}$$

Answer: $3.37 \times 10^{-19} \text{ J}$ ($\approx 2.1 \text{ eV}$)

(b)(iii) The 589.0 nm line is brighter because the transition producing it is more probable (higher transition probability). More electrons undergo this transition than the 589.6 nm transition.

Final answers

(a)(i) $P = 1.2 \times 10^{-2} \text{ W}$

**(a)(ii) Spot size constant $\Rightarrow$ irradiance
independent of distance $\Rightarrow$ unsuitable**

(b)(i) Electrons move to lower level

(b)(ii) $\Delta E = 3.37 \times 10^{-19} \text{ J}$ ($\approx 2.1 \text{ eV}$)

(b)(iii) 589.0 nm brighter: higher probability transition

Revision tips

- Irradiance $I = P/A$.

- Inverse square law requires spreading sphere of radius r ; collimator prevents this.
- Emission = electron moving to lower energy level; absorption = higher.
- Photon energy $E = hc/\lambda$.
- Relative brightness indicates relative transition probability.